AI Structural-Break Detector for Statistical Arbitrage

Econometric Relationship Engine

Mathematical and implementation specification

Complete workflow: inputs, transformations, integration tests, Engle-Granger estimation, cointegration vector, spread construction, rolling diagnostics, eligibility logic, event generation, and stored outputs

Role in the full AI system

This engine defines and monitors the historical relative-value relationship. It creates deterministic, auditable pair states and diagnostics for the downstream ML and agent layers. It does not decide whether a trade should be taken.

Core modelling choice

The baseline specification is a two-variable Engle-Granger relationship on log adjusted closing prices, estimated on a rolling 504-trading-day formation window and re-estimated monthly using only information available at the estimation date.

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Scope note

The main project overview left several econometric implementation details open. This document fixes a concrete baseline specification so the team can build reproducibly. Thresholds marked as configuration values may be tuned only on training/validation periods, never on the final test period.

1. Engine boundary and end-to-end workflow

1.1 Responsibility

The relationship engine receives cleaned daily price histories and a fixed pair registry. It determines whether each pair has a statistically defensible long-run relationship, estimates that relationship point-in-time, converts the fitted residual into a standardised divergence state, computes stability diagnostics, and determines whether the pair is ACTIVE, WATCH, or RETIRED.

1.2 Complete deterministic workflow

Step	Operation	Primary output
1	Load pair A/B price history and fixed pair orientation.	Aligned adjusted-close histories and data-quality flags.
2	Transform adjusted prices to log levels; compute log returns / first differences for diagnostics.	y_t, x_t, Δy_t, Δx_t.
3	Check integration order of each log-price series.	ADF/KPSS diagnostics; I(1)-compatibility flag.
4	Estimate long-run relation by OLS inside the 504-day formation window.	α-hat, β-hat, OLS residuals.
5	Run augmented Engle-Granger cointegration test.	Cointegration t-statistic and MacKinnon p-value.
6	Construct the residual spread and freeze relationship parameters for the signal month.	Cointegration vector, spread mean/std.
7	Estimate mean-reversion speed and stability statistics.	AR(1) ρ, half-life, beta drift, correlations, volatility, crossings.
8	Assign pair lifecycle state.	ACTIVE / WATCH / RETIRED.
9	During the signal month, compute daily spread and z-score using frozen parameters.	Daily pair-state table.
10	When an ACTIVE pair crosses the divergence threshold, create one event.	Event row and frozen feature timestamp.
11	Look forward only for label construction.	Success / failure / timeout label for ML training.

Separation from downstream AI

VIX, market returns, sector returns, NLP events, sentiment, SHAP, and agent findings are not regressors in the core cointegration equation. They are downstream ML/context features. This keeps the econometric relationship transparent and prevents the classical model from expanding into the main AI contribution.

2. Required data, variables, and transformations

2.1 Minimum data required by the relationship engine

Data item	Source table / field	Use in econometrics
Trading date	curated market prices: trading_date	Calendar alignment and point-in-time windows.
Adjusted close for leg A	adj_close for y_ticker	Log price y_t; corporate-action-consistent level.
Adjusted close for leg B	adj_close for x_ticker	Log price x_t.
Volume	volume for each leg	Not a regressor; optional liquidity gate and downstream feature.
Pair registry	pair_id, y_ticker, x_ticker, sector, active dates	Fixes economic pairing and OLS orientation ex ante.
Corporate-action audit	dividends, stock_splits	Quality control; adjusted prices remain modelling input.

2.2 Price-level transformation

Use strictly positive corporate-action-adjusted closing prices. For each security, transform the adjusted closing price to a natural logarithm. Log levels are the inputs to the long-run relationship; log differences are the daily continuously compounded returns used for integration and correlation diagnostics.

$$y_t = \ln(P_t^{A,\,adj}), \qquad x_t = \ln(P_t^{B,\,adj})$$

$$\Delta y_t = y_t - y_{t-1} = \ln\left(\frac{P_t^{A,\,adj}}{P_{t-1}^{A,\,adj}}\right), \qquad \Delta x_t = x_t - x_{t-1}$$

2.3 Calendar alignment

- Take the inner intersection of trading dates for the two securities.
- Never forward-fill missing equity prices across non-trading days or missing observations.
- Require a complete formation window after alignment; the default is 504 valid common trading days.
- A missing or non-positive adjusted close invalidates that observation before taking logs.

3. Pair orientation and pre-estimation eligibility

3.1 Pair orientation is fixed, not searched

Engle-Granger is not perfectly symmetric because the first-stage OLS regression treats one series as dependent. To avoid choosing the orientation that looks best in-sample, every pair receives a fixed y_ticker and x_ticker in the pair registry before backtesting. The same orientation is used throughout historical reconstruction, model training, test evaluation, and the UI.

Why this matters

Running both A-on-B and B-on-A and then choosing the stronger result would add an unrecorded model-selection step and increase data-snooping risk. A fixed orientation is reproducible and auditable.

3.2 Economic pre-screen

Gate	Baseline rule	Purpose
Economic similarity	Manually curated same-sector / same-sub-industry pair.	Avoid blind combinatorial pair mining.
History	At least 504 aligned observations before an estimation date.	Ensures full formation window.
Price validity	All modelling prices finite and > 0.	Required for log transformation.
Liquidity	Minimum volume/liquidity policy from data layer.	Prevents obviously non-tradable research cases.
Integration compatibility	Both log-price series plausibly I(1).	Required by Engle-Granger framework.

4. Integration-order diagnostics

4.1 Why the I(1) condition is checked

The augmented Engle-Granger test assumes the variables entering the cointegrating relationship are integrated of order one, I(1): their levels are non-stationary but their first differences are stationary. The engine therefore performs a diagnostic on each log price before using the pair as an ACTIVE cointegration candidate.

4.2 Augmented Dickey-Fuller test

$$\Delta q_t = a + \gamma q_{t-1} + \sum_{i=1}^{p} \phi_i \Delta q_{t-i} + \varepsilon_t$$

For q_t equal to y_t or x_t, the ADF null is a unit root. In levels, failure to reject is consistent with non-stationarity. In first differences, rejection of the unit-root null is consistent with stationarity. The baseline uses a constant and automatic lag selection by AIC.

4.3 KPSS cross-check

As a robustness diagnostic, KPSS reverses the null: stationarity is the null. Combining ADF and KPSS reduces reliance on a single test. KPSS is not required for the Engle-Granger computation itself but is useful for classifying ambiguous series.

Series state	ADF level	KPSS level	ADF first difference	Interpretation
Preferred I(1) pattern	Do not reject unit root	Reject stationarity	Reject unit root	Compatible with Engle-Granger.
Clearly I(0)	Reject unit root	Do not reject stationarity	-	Do not use as standard I(1)-I(1) pair.
Ambiguous	Tests conflict	Tests conflict	Investigate	WATCH / diagnostic flag; do not force a conclusion.
Potential I(2)	Do not reject	Reject	Do not reject	Incompatible with baseline engine.

Implementation rule

The integration diagnostics are a validity check, not a search for whichever p-value makes the pair eligible. Test specification and significance level are fixed globally.

5. Engle-Granger step 1: long-run OLS relationship

5.1 Cointegrating regression

Inside each formation window, estimate a constant-only long-run relationship between the two log adjusted prices:

$$y_t = \alpha + \beta x_t + u_t, \qquad t = T - W + 1, \ldots, T$$

Here W=504 trading days. The parameters are estimated by ordinary least squares. The intercept α captures the average level offset; β is the cointegrating coefficient used to construct the residual spread. With log prices, β is best interpreted as a long-run elasticity / relative exposure coefficient, not literally as the number of shares to short.

5.2 OLS objective

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha, \beta} \sum_{t=1}^{W} (y_t - \alpha - \beta x_t)^2$$

5.3 Estimated residuals

$$\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta} x_t$$

The residual is the candidate equilibrium error. Cointegration requires these residuals to be stationary even though y_t and x_t individually are non-stationary. A high OLS R-squared is not itself evidence of cointegration and is not used as the eligibility criterion.

No extra regressors in the core equation

SPY, sector ETFs, VIX, volume, valuation ratios, and NLP variables do not enter this OLS equation. The relationship engine intentionally remains a two-price model. Those other variables are used downstream as ML features or risk/context diagnostics.

6. Engle-Granger step 2: residual stationarity / cointegration test

6.1 Residual ADF concept

The second Engle-Granger step asks whether the estimated equilibrium error is stationary. Conceptually, the residual is tested with an augmented unit-root regression:

$$\Delta\hat{u}_t = \delta\hat{u}_{t-1} + \sum_{i=1}^{p} \psi_i \Delta\hat{u}_{t-i} + e_t$$

The null is no cointegration (residual contains a unit root). Rejection supports a stationary residual and therefore a cointegrating relationship. Because the residual is estimated in a first-stage regression, ordinary ADF critical values are not the correct Engle-Granger critical values.

6.2 Production test statistic and p-value

Use statsmodels.tsa.stattools.coint(y, x, trend="c", method="aeg", autolag="aic"). The function implements the augmented Engle-Granger two-step test and returns the cointegration test statistic together with MacKinnon approximate p-values/critical values appropriate to the cointegration setting. The baseline eligibility starting rule is p_coint < 0.05.

Output	Stored field	Use
Cointegration t-statistic	coint_tstat	Continuous relationship-quality feature and diagnostic.
MacKinnon p-value	coint_pvalue	Primary statistical eligibility gate.
Critical values	coint_crit_1pct/5pct/10pct	Audit and UI diagnostics.
Lag-selection configuration	eg_autolag	Reproducibility; baseline AIC.

Important implementation detail

Do not take the p-value returned by a generic adfuller() call on the first-stage residuals and treat it as the Engle-Granger cointegration p-value. The production gate uses coint(), which applies the appropriate MacKinnon cointegration distribution.

7. Cointegration vector, equilibrium spread, and hedge coefficient

7.1 Cointegration vector

The fitted long-run relation can be written as a stationary linear combination. Ignoring the intercept for vector notation, the two-asset cointegration vector is proportional to (1, -β-hat). The intercept is stored separately.

$$\mathbf{b} = (1, \ -\hat{\beta}), \qquad s_t = y_t - \hat{\alpha} - \hat{\beta} x_t$$

7.2 Spread interpretation

- s_t > 0: leg A is high relative to the fitted relationship with leg B.
- s_t < 0: leg A is low relative to the fitted relationship.
- s_t near the formation equilibrium: the pair is close to its estimated long-run relation.
- The spread is a statistical state variable. It is not itself a guaranteed arbitrage profit.

7.3 Relationship coefficient versus trading weights

The econometric engine stores β-hat as the relationship coefficient. The backtester may use it to form a beta-scaled relative return, but actual executable share quantities depend on current prices, capital, borrow constraints, and transaction costs. Those implementation choices belong to the backtesting/execution layer, not to the cointegration test.

$$\Delta s_t \approx r_t^A - \hat{\beta}\, r_t^B$$

8. Point-in-time rolling estimation design

8.1 Formation and signal periods

The relationship is re-estimated monthly. On each estimation date T, the engine uses only the previous 504 common trading observations. The resulting parameters become effective from the next trading session and remain frozen until the next scheduled re-estimation.

$$\mathcal{W}_T = \{T - W + 1, \ldots, T\}, \qquad W = 504$$

8.2 Why parameters are frozen between re-estimations

- A divergence should be measured against a relationship known before the signal, not a regression that immediately adapts to the divergence.
- Freezing α-hat and β-hat prevents the model from partially fitting away the event it is supposed to detect.
- Monthly snapshots make historical reconstruction and app displays deterministic and easy to audit.

8.3 Effective-time convention

Timestamp	Rule
Estimation close T	Formation window ends at T close. Fit and diagnostics are computed after the close.
Effective start	The new snapshot is first eligible for signals on the next trading day T+1.
During signal month	Use the same α-hat, β-hat, formation spread mean/std, and status until the next effective snapshot.
Historical backtest	Each event joins to the latest snapshot with effective_start <= event_date; never to a future monthly estimate.

9. Spread standardisation and daily pair state

9.1 Formation spread moments

From the in-window OLS residuals, store the formation spread mean and standard deviation. With an intercept, the residual mean is numerically close to zero, but it is stored explicitly for transparency and reproducibility.

$$\hat{\mu}_s = \frac{1}{W} \sum_{t \in \mathcal{W}_T} \hat{u}_t, \qquad \hat{\sigma}_s = \sqrt{\frac{1}{W-1} \sum_{t \in \mathcal{W}_T} (\hat{u}_t - \hat{\mu}_s)^2}$$

9.2 Out-of-sample daily spread and z-score

For any signal date τ after the estimation date and before the next re-estimation, calculate the spread with the frozen coefficients and standardise it with the frozen formation moments:

$$s_\tau = y_\tau - \hat{\alpha}_T - \hat{\beta}_T x_\tau, \qquad z_\tau = \frac{s_\tau - \hat{\mu}_{s,T}}{\hat{\sigma}_{s,T}}$$

Why the standardisation is frozen

Recomputing the mean and variance with future signal-period observations would allow the reference distribution to adapt to the divergence. The baseline therefore uses formation-window moments until the next monthly estimation snapshot.

9.3 Daily state variables

Variable	Definition / transformation
spread	Current s_tau from frozen α-hat and β-hat.
z_score	Current spread in formation standard deviations.
spread_velocity_1d	s_tau - s_(tau-1).
z_velocity_1d	z_tau - z_(tau-1).
abs_z	|z_tau|.
days_since_zero_cross	Trading days since spread last changed sign.
pair_status	ACTIVE / WATCH / RETIRED from the latest monthly snapshot.

10. Mean-reversion speed and half-life

10.1 AR(1) spread model

Half-life is estimated as a descriptive relationship-quality statistic using the formation-window spread. Fit a first-order autoregression:

$$s_t = c + \rho s_{t-1} + \eta_t$$

For a stable positive autoregressive coefficient $0 < \rho < 1$, the implied half-life is the number of trading days required for a deviation to decay by one half under the AR(1) approximation:

$$h_{1/2} = -\frac{\ln 2}{\ln \rho}$$

10.2 Validity rules

Case	Treatment
$0 < \rho < 1$	Store finite half-life.
$\rho >= 1$	No mean-reverting AR(1) interpretation; half-life set null/invalid and relationship flagged.
$\rho <= 0$	Oscillatory / very fast sign switching; do not force half-life formula; flag for review.
Very long half-life	Relationship may be statistically cointegrated but too slow for the 10-day event horizon; retain as feature/diagnostic rather than silently excluding unless policy specifies a gate.

10.3 Equivalent error-correction representation

$$\Delta s_t = \kappa + \phi s_{t-1} + \eta_t, \qquad \phi = \rho - 1$$

Negative φ corresponds to error correction. The project can store φ as an alternative mean-reversion-speed feature. A full VECM is not required for the core two-asset engine; the simple AR(1) spread diagnostic is sufficient and keeps the econometrics supporting rather than dominant.

11. Relationship stability diagnostics

11.1 Core diagnostics computed each monthly snapshot

Diagnostic	Mathematical definition / rule	Role
Cointegration history	Current and previous coint_tstat / p-value.	Primary validity history.
Return correlation 20d / 60d	Corr(Δy, Δx) over trailing 20/60 common days.	Short/medium co-movement context; not a substitute for cointegration.
Beta drift	|β_T - β_prev| / max(|β_prev|, ε).	Detect changing long-run exposure.
Spread volatility	Std(s) over trailing 20d; ratio to formation σ_s.	Detect sudden instability.
Zero crossings	Count of sign changes in s over trailing 252d.	Empirical evidence that equilibrium is revisited.
Recency of crossing	Days since latest sign change.	Identifies spreads drifting one-sided for long periods.
Half-life	AR(1) implied h_1/2.	Mean-reversion speed feature.
Formation residual range	max(z_in) - min(z_in).	Scale / tail diagnostic.

11.2 Optional structural-stability test

CUSUM can be computed on the first-stage OLS residuals as a secondary warning flag. Its null is parameter stability. This is not the project's main structural-break detector and should not override the AI evidence layer; it is an econometric diagnostic only.

Implementation: statsmodels.stats.diagnostic.breaks_cusumolsresid(resid, ddof=2). Store the statistic, p-value, and 1%/5%/10% critical values. A low p-value raises a stability warning.

Avoid diagnostic overload

Do not add every possible time-series test. The engine should expose a small set of interpretable, pre-specified statistics that are either used as gates, ML features, or visible diagnostics. Tests with no downstream role increase complexity without improving the research design.

12. Pair lifecycle and gating rules

12.1 Monthly status assignment

Status	Core rule	New divergence events?
ACTIVE	Full history; I(1)-compatible; coint p-value below policy threshold; no hard data-quality failure.	Yes.
WATCH	One monthly snapshot fails cointegration or a material stability/data-quality warning is triggered.	No in baseline; UI may display relationship deterioration.
RETIRED	Two consecutive monthly invalid snapshots, delisting/major transformation, or persistent unusable data/liquidity.	No.

12.2 Default hard versus soft gates

Type	Examples	Effect
Hard gate	Insufficient 504-day history; invalid prices; series clearly not I(1); coint p >= 0.05; missing current price.	Pair cannot be ACTIVE.
Soft diagnostic	Large beta drift; long half-life; elevated spread volatility; low crossing frequency; CUSUM warning.	Stored and surfaced; can move pair to WATCH by pre-specified policy or feed downstream ML.
Manual lifecycle flag	Merger completion, delisting, ticker replacement, permanent corporate restructuring.	RETIRED with reason code and effective date.

No p-value shopping

The 5% threshold is a baseline configuration. If the team compares 1%, 5%, and 10%, the choice is made on training/validation only and locked before the final test period.

13. Divergence-event generation and econometric labels

13.1 Event trigger

The relationship engine creates an event only when an ACTIVE pair enters the divergence region from inside it. This avoids treating every day of the same prolonged divergence as a new independent observation.

$$E_t = 1[|z_t| \geq z_{entry}, \quad |z_{t-1}| < z_{entry}]$$

Baseline z_entry = 2.0. Store direction d_t = sign(z_t): +1 means A is high relative to B; -1 means A is low relative to B.

13.2 Event resolution and target label

$$Y_i = 1 \quad \text{if} \quad \min_{1 \leq h \leq H} |z_{t_i + h}| \leq z_{exit}$$

Baseline: z_exit = 0.5, H = 10 trading days. The adverse boundary is direction-specific: for d=+1, failure occurs if z reaches or exceeds +3.0 before convergence; for d=-1, failure occurs if z reaches or falls below -3.0. If neither condition occurs by H, label failure/timeout according to the project target definition.

Field	Value stored at event time
event_id	Unique immutable event key.
pair_id	Pair registry key.
event_date	End-of-day signal date.
snapshot_id	Latest effective monthly econometric snapshot.
z_entry	Observed z-score at trigger.
direction	+1 or -1.
alpha / beta	Frozen relationship coefficients.
coint_tstat / pvalue	Relationship validity diagnostics known at event time.
half_life / beta_drift / correlations	Frozen or trailing diagnostics known at event time.
label fields	Written only by the later label-construction job; never available to feature generation.

14. Outputs passed to ML and the UI

14.1 Relationship-quality ML features

Feature	Type	Point-in-time meaning
coint_tstat	continuous	Strength of current Engle-Granger rejection.
coint_pvalue	continuous	Cointegration test p-value.
beta	continuous	Current long-run relative coefficient.
beta_drift_pct	continuous	Change in coefficient from prior monthly snapshot.
half_life_days	continuous / nullable	AR(1) mean-reversion speed.
corr_20d / corr_60d	continuous	Recent return co-movement.
spread_vol_20d_ratio	continuous	Recent spread volatility relative to formation scale.
zero_crossings_252d	integer	Recent equilibrium-crossing frequency.
days_since_zero_cross	integer	Recency of last crossing.
relationship_age_months	integer	Number of consecutive valid monthly snapshots.
integration_warning	categorical/binary	Ambiguous I(1) diagnostics.
cusum_warning	binary optional	Secondary parameter-stability warning.

14.2 Spread-state ML features

Feature	Definition
z_score	Current standardised divergence.
abs_z	Magnitude of divergence.
spread_velocity_1d	One-day change in raw spread.
z_velocity_1d	One-day change in z-score.
distance_to_exit	abs_z - z_exit.
distance_to_adverse	z_stop - abs_z in the event direction.
prior_crossing_count	Number of previous qualifying divergence events for the pair in a trailing window, computed only from past data.

What is not passed from this engine

The engine does not create market-context features such as VIX, market/sector returns, abnormal volume, days-to-earnings, news sentiment, event materiality, or agent conclusions. Those are constructed by other layers and joined by event_id / event timestamp.

15. Exact implementation function map and tech stack

15.1 Python stack

Component	Library / function	Purpose
Tabular/time alignment	pandas	Date-index joins, rolling windows, group operations.
Numeric transforms	numpy	log, differencing, vectorised calculations.
Long-run OLS	statsmodels.api.OLS + statsmodels.tools.add_constant	Estimate α and β and retain residuals.
Engle-Granger test	statsmodels.tsa.stattools.coint	Primary no-cointegration test with MacKinnon p-value.
ADF diagnostics	statsmodels.tsa.stattools.adfuller	Integration-order diagnostics on price levels/differences.
KPSS diagnostics	statsmodels.tsa.stattools.kpss	Stationarity-null cross-check.
Optional CUSUM	statsmodels.stats.diagnostic.breaks_cusumolsresid	Secondary parameter-stability diagnostic.
Persistence	duckdb + parquet	Store monthly snapshots, daily states, events and audit lineage.

15.2 Application-level function responsibilities

Function	Inputs	Returns
load_pair_window	pair_id, estimation_date, snapshot_id	Aligned 504-day A/B adjusted-price frame.
transform_pair_prices	aligned price frame	y, x, Δy, Δx and validation flags.
test_integration_order	y, x	ADF/KPSS result object and compatibility flags.
fit_cointegrating_regression	y, x	α, β, residual vector, OLS diagnostics.
run_engle_granger	y, x	coint t-stat, p-value, critical values, test settings.
estimate_half_life	formation spread	AR(1) ρ, φ, half-life and validity flag.
compute_stability_metrics	current/prior snapshots + price frame	correlations, beta drift, spread-vol ratio, crossings, optional CUSUM.
assign_pair_status	all tests + lifecycle history	ACTIVE/WATCH/RETIRED and reason codes.
build_daily_pair_state	snapshot + subsequent daily prices	spread, z-score, velocity fields per date.
detect_divergence_events	daily state + unresolved-event state	non-overlapping event records.
label_event	future z path, fixed horizon	success/failure/timeout and resolution metadata.
persist_relationship_snapshot	result object + lineage IDs	DuckDB monthly pair snapshot row(s).

No hidden side effects

Each function should be deterministic for a fixed data_snapshot_id and configuration version. Retrieval from external providers belongs to the data-ingestion layer, not inside econometric functions.

16. Persistent schemas and data lineage

16.1 Monthly relationship snapshot

Field group	Required fields
Identity	pair_id, y_ticker, x_ticker, estimation_date, effective_start, effective_end, relationship_spec_version.
Window	window_start, window_end, n_obs, formation_window_days.
Integration	y_adf_level_p, y_adf_diff_p, x_adf_level_p, x_adf_diff_p, optional KPSS fields, integration_flag.
Engle-Granger	alpha, beta, coint_tstat, coint_pvalue, critical values, trend_spec, autolag_spec.
Spread scale	spread_mean, spread_std.
Mean reversion	ar1_rho, error_correction_phi, half_life_days, half_life_valid.
Stability	corr_20d, corr_60d, beta_drift_pct, spread_vol_20d, spread_vol_ratio, zero_crossings_252d, days_since_zero_cross, optional cusum_pvalue.
Lifecycle	pair_status, status_reason, consecutive_invalid_snapshots.
Lineage	data_snapshot_id, source_price_snapshot_ids, config_hash, created_at_utc.

16.2 Daily pair state

Field	Purpose
pair_id, trading_date, relationship_snapshot_id	Join keys.
alpha, beta	Frozen coefficients used that day.
spread, z_score, abs_z	Current relationship state.
spread_velocity_1d, z_velocity_1d	Short-run divergence dynamics.
pair_status	Eligibility inherited from snapshot.
data_available_at	Point-in-time availability boundary.
data_snapshot_id	Reproducibility.

17. Leakage controls, validation, and failure handling

17.1 Non-negotiable temporal controls

- Formation-window observations must be on or before the estimation date.
- A monthly relationship snapshot becomes effective only after it is estimated; historical events cannot join to later snapshots.
- No future price is used in spread scaling, half-life estimation, or event features.
- Future z-scores are read only by the label-construction job after the event row is frozen.
- No threshold, pair orientation, formation-window length, or cointegration significance rule may be changed after reviewing final test results.

17.2 Numerical/data failure conditions

Failure	Handling
Near-constant or invalid price series	Reject window; data-quality reason code.
Perfect/near-perfect collinearity	Store warning; statsmodels coint may return extreme statistic; review raw data for duplication.
spread_std <= epsilon	Reject snapshot; z-score undefined.
Insufficient common dates	No snapshot; pair WATCH/RETIRED depending persistence.
Missing current signal-day price	No daily state/event for that date; no forward fill.
Half-life invalid	Store null + flag; do not fabricate a value.
OLS / statistical test exception	Persist failed job record with pair/date/config; do not silently skip.

17.3 Unit and integration tests for code

- Synthetic cointegrated series should be classified as cointegrated at a high rate under the fixed specification.
- Independent random walks should usually fail the cointegration gate.
- No feature row may contain dates later than its event timestamp.
- Re-running the same snapshot/configuration must reproduce identical α, β, z-scores, events, and labels.
- Changing the pair orientation must change the output in a documented way; production orientation remains fixed by registry.

18. Recommended baseline configuration and sensitivity grid

18.1 Locked starting configuration

Parameter	Baseline value	Why
Price input	log adjusted close	Corporate-action-consistent level relation.
Pair model	Engle-Granger, constant only	Simple, interpretable two-asset long-run relation.
Formation window	504 trading days	~2 years; balances stability and adaptation.
Re-estimation	Monthly, effective next session	Point-in-time and computationally manageable.
EG autolag	AIC	Standard automatic lag selection.
Cointegration alpha	5%	Initial eligibility threshold.
Entry z	2.0	Meaningful statistical divergence.
Exit z	0.5	Defines successful convergence.
Adverse z	3.0 in event direction	Failure boundary.
Label horizon	10 trading days	Short-term statistical-arbitrage horizon.
WATCH rule	1 failed monthly validity snapshot	Conservative suppression of new signals.
RETIRED rule	2 consecutive invalid monthly snapshots or manual terminal reason	Removes persistently broken relationships.
CUSUM	Diagnostic only	Does not duplicate AI structural-break decision.

18.2 Sensitivity analysis - validation only

Dimension	Permitted validation grid
Formation window	252 / 504 / 756 trading days.
Cointegration threshold	1% / 5% / 10%.
Entry threshold	1.5 / 2.0 / 2.5 standard deviations.
Exit threshold	0.0 / 0.5 / 1.0 standard deviations.
Label horizon	5 / 10 / 20 trading days.
Adverse boundary	2.5 / 3.0 / 3.5 standard deviations.
Re-estimation cadence	Monthly baseline; weekly only as sensitivity if computation permits.

Model-selection rule

The final configuration is selected using only training and validation periods and then frozen. The final test period is opened once. Sensitivity results are reported transparently rather than used to cherry-pick the best historical outcome.

Appendix A. Formula reference

Object	Formula / definition
Log prices	y_t = ln(P_A,adj,t), x_t = ln(P_B,adj,t).
Log returns	Δy_t = y_t - y_(t-1), Δx_t = x_t - x_(t-1).
Long-run relation	y_t = α + β x_t + u_t.
OLS estimate	Minimise Σ(y_t - α - βx_t)^2.
Cointegration vector	b = (1, -β-hat); intercept stored separately.
Spread	s_t = y_t - α-hat - β-hat x_t.
Z-score	z_t = (s_t - μ_s) / σ_s, with μ_s and σ_s frozen from formation window.
AR(1)	s_t = c + ρ s_(t-1) + η_t.
Half-life	-ln(2)/ln(ρ), only when 0 < ρ < 1.
Error-correction coefficient	φ = ρ - 1 in Δs_t = κ + φ s_(t-1) + η_t.
Beta drift	|β_T - β_prev| / max(|β_prev|, ε).
Event trigger	|z_t| >= z_entry and |z_(t-1)| < z_entry and pair ACTIVE.
Success label	Reach |z| <= z_exit within H before hitting adverse boundary.

Appendix B. Interpretation of every econometric output

Output	Interpretation	Common misuse to avoid
α-hat	Level offset in long-run log-price relation.	Do not interpret as expected return.
β-hat	Relative long-run coefficient / elasticity.	Do not call it an executable share ratio without mapping to notionals.
coint_tstat	Evidence against no-cointegration; more negative generally stronger.	Do not compare to ordinary t-test critical values.
coint_pvalue	MacKinnon probability measure for the no-cointegration null.	Do not treat p<0.05 as proof relationship will persist forever.
spread	Estimated equilibrium error.	Do not confuse a large spread with guaranteed mispricing.
z_score	Spread measured in formation standard deviations.	Do not re-standardise using future information.
ρ	Persistence of AR(1) spread approximation.	Do not assume the AR(1) is a full structural model.
half_life	Approximate decay speed conditional on AR(1) validity.	Do not use as a guaranteed exit time.
corr_20d/60d	Short-run co-movement of returns.	Correlation is not cointegration.
beta_drift	Instability in estimated long-run exposure.	A stable beta alone does not prove cointegration.
CUSUM p-value	Secondary parameter-stability warning.	Do not let it replace the AI structural-break evidence layer.
pair_status	Operational eligibility state under policy.	It is not a trade recommendation.

References

Project documents

- FINS5557 Track A project concept: AI Structural-Break Detector for Statistical Arbitrage.
- AI Structural-Break Detector - Complete Project Overview.
- AI Structural-Break Detector - Data Architecture and Source Strategy.

Econometric / implementation references

- Engle, R. F. and Granger, C. W. J. (1987). Co-integration and Error Correction: Representation, Estimation, and Testing. Econometrica 55(2), 251-276.
- MacKinnon, J. G. (1994; 2010). Approximate distribution functions / critical values for cointegration and unit-root tests.
- statsmodels documentation (0.15.x): tsa.stattools.coint, adfuller, kpss; api.OLS; stats.diagnostic.breaks_cusumolsresid.

Implementation interpretation

The econometric engine establishes a point-in-time statistical relationship and produces transparent diagnostics. The custom ML model and agentic evidence system remain responsible for learning which divergence events are attractive and which relationships may have economically broken.